

Fourier Tower Transforms: Worked Examples for Finite Groups and Quantum Circuits

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Abstract

This article gives concrete worked examples of Fourier Tower transforms for solvable finite groups and symmetry-adapted quantum circuits.

The purpose is to illustrate explicitly how Fourier Tower ideas work in practice. We study examples including S_3 , S_4 , the dihedral group D_4 , the quaternion group Q_8 , the finite Heisenberg group H_p , and quantum gates such as Toffoli and Fredkin.

In each case, we identify the relevant tower, compute the local Fourier layers, describe the coupling operators, and give the associated quantum circuit.

The examples show concretely how nonabelian structure can be understood as an interaction between local abelian Fourier sectors, either through controlled permutations or through cocycle-induced phase twisting.

1 Preliminaries

Let G be a finite solvable group. A Fourier Tower starts from a normal series

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G,$$

whose quotient groups

$$Q_i = G_i/G_{i-1}$$

are abelian.

Each quotient layer has a Hilbert space

$$\mathcal{H}_i = \mathbb{C}[Q_i].$$

Since Q_i is abelian, the local Fourier transform is the ordinary abelian Fourier transform

$$F_{Q_i} : \mathbb{C}[Q_i] \rightarrow \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi.$$

For a cyclic group $\mathbb{Z}_n$, this is

$$F_n|x\rangle = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} e^{2\pi i x m/n} |m\rangle.$$

The local Fourier layer is

$$F_{\text{loc}} = \bigotimes_i F_{Q_i}.$$

The nonabelian structure appears through conjugation. If a higher quotient layer acts on a lower one by

$$\sigma_y(x) = yxy^{-1},$$

then the corresponding controlled conjugation gate is

$$C_i = \sum_y |y\rangle\langle y| \otimes P_y,$$

where

$$P_y|x\rangle = |\sigma_y(x)\rangle.$$

Thus the Fourier Tower transform has the form

$$F_{\mathcal{T}} = \left(\prod_i C_i \right) \left(\bigotimes_i F_{Q_i} \right).$$

The guiding principle is

nonabelian structure = interaction between local Fourier sectors.

2 Worked Example: The Fourier Tower for S_3

2.1 Solvable Structure

The group S_3 has the presentation

$$S_3 = \langle r, s \mid r^3 = e, s^2 = e, srs^{-1} = r^{-1} \rangle.$$

It admits the solvable tower

$$\{e\} \triangleleft A_3 \triangleleft S_3,$$

where

$$A_3 = \langle r \rangle \cong \mathbb{Z}_3$$

and

$$S_3/A_3 \cong \mathbb{Z}_2.$$

Therefore the Fourier Tower has two quotient layers:

$$Q_1 \cong \mathbb{Z}_3, \quad Q_2 \cong \mathbb{Z}_2.$$

The Hilbert space is

$$\mathcal{H} = \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_3].$$

A basis vector is written

$$|a, b\rangle, \quad a \in \mathbb{Z}_2, \quad b \in \mathbb{Z}_3.$$

Here a records the quotient layer $\mathbb{Z}_2$, while b records the normal subgroup layer $\mathbb{Z}_3$.

2.2 Local Fourier Layers

The Fourier transform on $\mathbb{Z}_2$ is

$$F_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Let

$$\omega = e^{2\pi i/3}.$$

The Fourier transform on $\mathbb{Z}_3$ is

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}.$$

The uncoupled local Fourier layer is

$$F_{\text{loc}} = F_2 \otimes F_3.$$

This operator performs Fourier analysis separately on the two abelian layers.

2.3 Conjugation Coupling

The nonabelian relation is

$$srs^{-1} = r^{-1}.$$

On the $\mathbb{Z}_3$ layer, this means

$$b \mapsto -b.$$

Thus

$$0 \mapsto 0, \quad 1 \mapsto 2, \quad 2 \mapsto 1.$$

The permutation matrix is

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

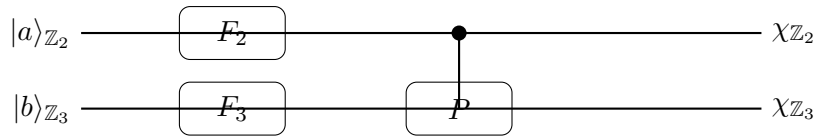
The controlled conjugation operator is

$$C = |0\rangle\langle 0| \otimes I_3 + |1\rangle\langle 1| \otimes P.$$

Therefore the Fourier Tower transform is

$$F_{\mathcal{T}} = C(F_2 \otimes F_3).$$

2.4 Circuit for S_3



The upper register controls whether the lower $\mathbb{Z}_3$ Fourier modes are inverted. The two nontrivial characters of $\mathbb{Z}_3$ are exchanged:

$$\chi_1 \leftrightarrow \chi_2.$$

3 Worked Example: The Fourier Tower for S_4

3.1 Solvable Structure

The group S_4 admits the normal series

$$\{e\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4,$$

where

$$V_4 = \{e, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The quotient layers are

$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

$$A_4/V_4 \cong \mathbb{Z}_3,$$

and

$$S_4/A_4 \cong \mathbb{Z}_2.$$

Thus the tower has three main layers:

$$\mathbb{Z}_2, \quad \mathbb{Z}_3, \quad V_4.$$

The Hilbert space is

$$\mathcal{H} = \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_3] \otimes \mathbb{C}[V_4].$$

Since

$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

we may also write

$$\mathcal{H} \cong \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_3] \otimes \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2].$$

3.2 Local Fourier Layers

The top $\mathbb{Z}_2$ layer has Fourier transform

$$F_2 = H.$$

The middle $\mathbb{Z}_3$ layer has Fourier transform

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}.$$

The bottom layer is

$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Therefore

$$F_{V_4} = F_2 \otimes F_2.$$

Explicitly,

$$F_{V_4} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Thus the local Fourier layer is

$$F_{\text{loc}} = F_2 \otimes F_3 \otimes F_{V_4}.$$

3.3 Conjugation Couplings

Inside A_4 , the quotient

$$A_4/V_4 \cong \mathbb{Z}_3$$

acts cyclically on the three nontrivial elements of V_4 :

$$(12)(34) \mapsto (13)(24) \mapsto (14)(23) \mapsto (12)(34).$$

Thus the $\mathbb{Z}_3$ layer cyclically permutes the nontrivial Fourier sectors of V_4 .

Let P_3 denote this cyclic permutation operator. Then the first controlled coupling is

$$C_1 = \sum_{b \in \mathbb{Z}_3} |b\rangle\langle b| \otimes P_3^b.$$

The top quotient

$$S_4/A_4 \cong \mathbb{Z}_2$$

acts on the $\mathbb{Z}_3$ layer by inversion:

$$b \mapsto -b.$$

This is represented by

$$P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The second controlled coupling is

$$C_2 = |0\rangle\langle 0| \otimes I_3 + |1\rangle\langle 1| \otimes P_2.$$

Therefore the Fourier Tower transform for S_4 is

$$F_{\mathcal{T}} = (C_2 C_1)(F_2 \otimes F_3 \otimes F_{V_4}).$$

3.4 Circuit for S_4

4 Worked Example: The Fourier Tower for D_4

The dihedral group D_4 has presentation

$$D_4 = \langle r, s \mid r^4 = e, s^2 = e, srs^{-1} = r^{-1} \rangle.$$

A solvable tower is

$$\{e\} \triangleleft \langle r^2 \rangle \triangleleft \langle r \rangle \triangleleft D_4.$$

The quotient layers are all isomorphic to $\mathbb{Z}_2$:

$$Q_1 \cong \mathbb{Z}_2, \quad Q_2 \cong \mathbb{Z}_2, \quad Q_3 \cong \mathbb{Z}_2.$$

Hence

$$\mathcal{H} = \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2].$$

The local Fourier transform is

$$F_{\text{loc}} = H \otimes H \otimes H.$$

The nonabelian relation

$$srs^{-1} = r^{-1}$$

means that the reflection layer controls inversion on the rotational layers.

Thus

$$F_{\mathcal{T}} = C_s(H \otimes H \otimes H).$$

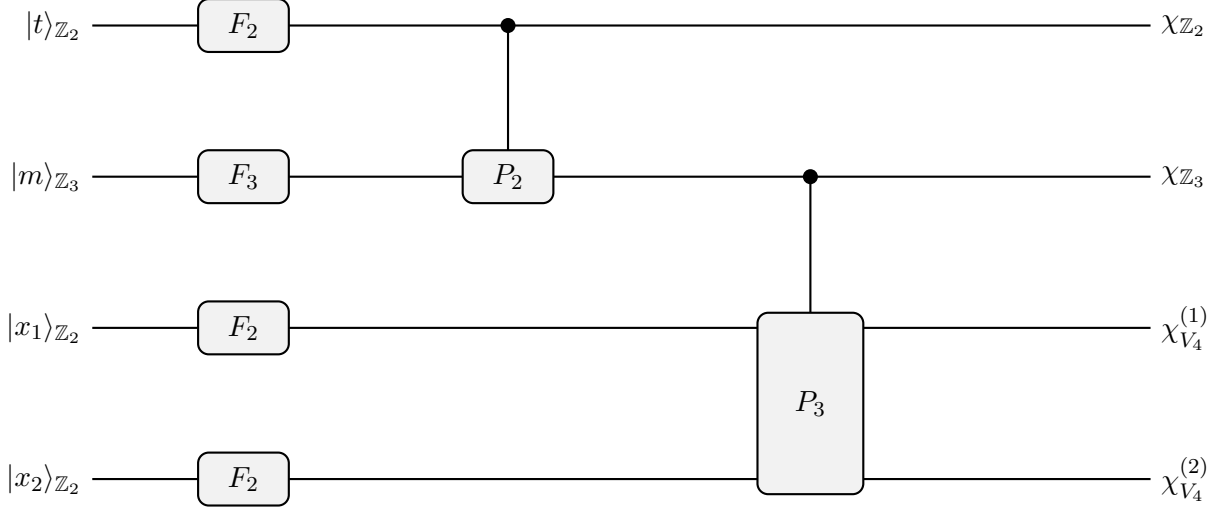


Figure 1: Fourier Tower circuit for S_4 . The gate P_2 acts on the $\mathbb{Z}_3$ wire and is controlled by the top $\mathbb{Z}_2$ layer. The gate P_3 acts on the two wires encoding $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and is controlled by the $\mathbb{Z}_3$ layer.

5 Worked Example: Cocycle-Twisted Fourier Layer

Consider

$$Q = \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The ordinary characters are

$$\chi_{ab}(x, y) = (-1)^{ax+by}.$$

A cocycle-twisted Fourier mode is written as

$$|\chi_{ab}^\omega\rangle = \frac{1}{2} \sum_{(x,y) \in Q} \chi_{ab}(x, y) \theta_\omega(x, y) |(x, y)\rangle.$$

Here

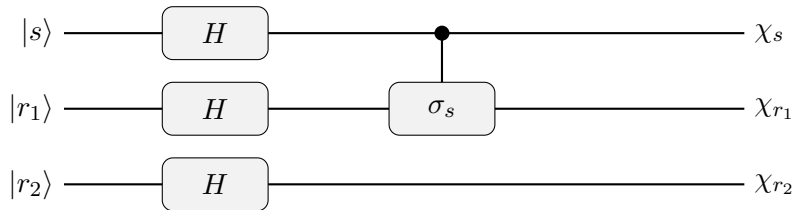
$$\theta_\omega(x, y) \in U(1)$$

is a phase determined by the cocycle.

The corresponding twisted Fourier transform is

$$F_Q^\omega |x, y\rangle = \frac{1}{2} \sum_{a, b \in \mathbb{Z}_2} (-1)^{ax+by} \theta_\omega(x, y) |a, b\rangle.$$

5.1 Circuit for D_4



6 Worked Example: The Fourier Tower for Q_8

The quaternion group is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad ji = -k.$$

Let

$$z = -1.$$

Then

$$Z(Q_8) = \{1, z\} \cong \mathbb{Z}_2.$$

A solvable tower is

$$\{1\} \triangleleft \langle z \rangle \triangleleft \langle i \rangle \triangleleft Q_8.$$

The quotient layers are all isomorphic to $\mathbb{Z}_2$.

Thus

$$\mathcal{H} = \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2].$$

Every element may be written as

$$j^c i^b z^a, \quad a, b, c \in \{0, 1\}.$$

The local Fourier transform is

$$F_{\text{loc}} = H \otimes H \otimes H.$$

The relations

$$i^2 = z, \quad j^2 = z, \quad ji = zij$$

show that multiplication produces extra central factors.

Thus the nonabelian structure of Q_8 is encoded by a central cocycle.

In the Fourier basis of the central layer, multiplication by z produces a phase:

$$|0\rangle \mapsto |0\rangle, \quad |1\rangle \mapsto -|1\rangle.$$

Therefore the cocycle factors become controlled phase gates.

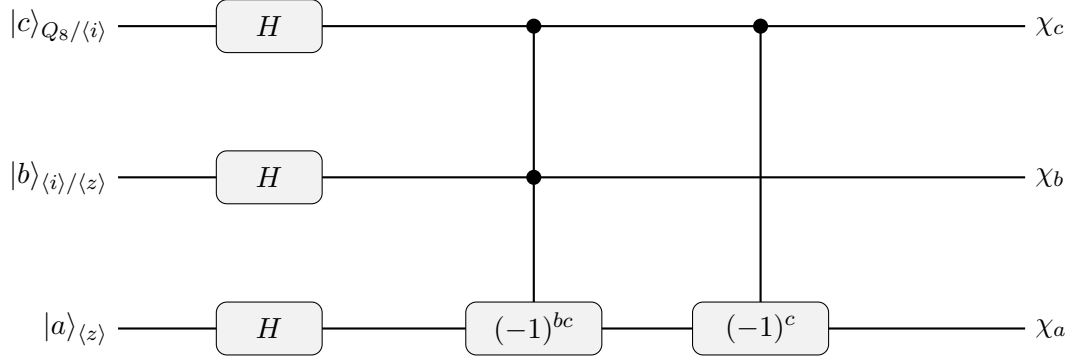
The Fourier Tower transform for Q_8 may be written schematically as

$$F_{\mathcal{T}}^{Q_8} = C_{\omega}(H \otimes H \otimes H).$$

6.1 Circuit for Q_8

The Fourier Tower circuit for Q_8 consists of:

1. three local Hadamard transforms,
2. a cocycle-controlled phase gate coming from $ji = zij$,
3. and a central phase gate coming from $j^2 = z$.



The first phase gate represents the cocycle contribution coming from

$$ji = zij.$$

The second phase gate represents the central contribution coming from

$$j^2 = z.$$

Thus the Q_8 Fourier Tower shows that nonabelianity can appear as a central phase interaction rather than as a permutation of Fourier modes.

7 Worked Example: Fourier Tower for the Finite Heisenberg Group

The finite Heisenberg group is

$$H_p = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{F}_p \right\}.$$

Equivalently, elements are triples

$$(a, b, c), \quad a, b, c \in \mathbb{F}_p.$$

The multiplication is

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab').$$

The center is

$$Z(H_p) = \{(0, 0, c) : c \in \mathbb{F}_p\} \cong \mathbb{Z}_p.$$

Moreover,

$$H_p/Z(H_p) \cong \mathbb{Z}_p \times \mathbb{Z}_p.$$

Thus H_p is a central extension

$$1 \rightarrow \mathbb{Z}_p \rightarrow H_p \rightarrow \mathbb{Z}_p \times \mathbb{Z}_p \rightarrow 1.$$

The cocycle is

$$\omega((a, b), (a', b')) = ab'.$$

The local Fourier transform is

$$F_{\text{loc}} = F_p \otimes F_p \otimes F_p.$$

The cocycle produces a controlled phase interaction

$$|a\rangle|b\rangle|c\rangle \mapsto \omega_p^{ab}|a\rangle|b\rangle|c\rangle,$$

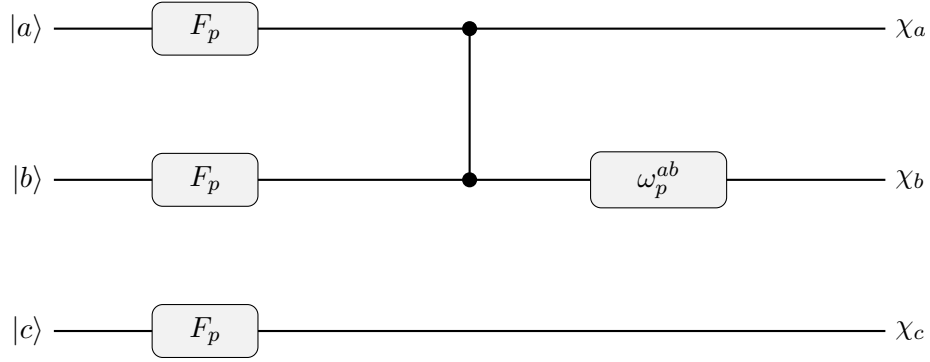
where

$$\omega_p = e^{2\pi i/p}.$$

Thus the Fourier Tower transform is

$$F_{\mathcal{T}} = C_{\omega}(F_p \otimes F_p \otimes F_p).$$

7.1 Circuit for H_p



7.2 Quantum and Physical Meaning

The finite Heisenberg group is the discrete analogue of the Heisenberg symmetry of quantum mechanics.

The quotient coordinates (a, b) behave locally like abelian translation directions, while the central coordinate c stores the phase correction.

The multiplication law

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab')$$

shows that the term

$$ab'$$

is the cocycle responsible for noncommutativity.

After Fourier transformation of the central layer, the translation

$$c \mapsto c + ab'$$

becomes a phase factor

$$\omega_p^{ab'}.$$

Thus the Fourier Tower converts the central cocycle into a controlled phase interaction.

This connects the finite Heisenberg example with generalized Pauli operators

$$X|x\rangle = |x+1\rangle, \quad Z|x\rangle = \omega_p^x|x\rangle,$$

which satisfy

$$ZX = \omega_p XZ.$$

Therefore the Heisenberg Fourier Tower is naturally related to stabilizer quantum mechanics, Clifford circuits, and finite-dimensional quantum phase space.

8 Fourier Tower Representation of the Toffoli Gate

The Toffoli gate is

$$U = \text{CCX},$$

defined by

$$\text{CCX}|a, b, c\rangle = |a, b, c \oplus ab\rangle.$$

The first two qubits are control layers, while the third qubit is the target layer.
The target register carries the abelian group structure

$$\mathbb{Z}_2.$$

Let

$$X|c\rangle = |c \oplus 1\rangle.$$

Then

$$U = |0, 0\rangle\langle 0, 0| \otimes I + |0, 1\rangle\langle 0, 1| \otimes I + |1, 0\rangle\langle 1, 0| \otimes I + |1, 1\rangle\langle 1, 1| \otimes X.$$

The Fourier transform on the target layer is

$$F_{\text{loc}} = I \otimes I \otimes H.$$

Using

$$HXH = Z,$$

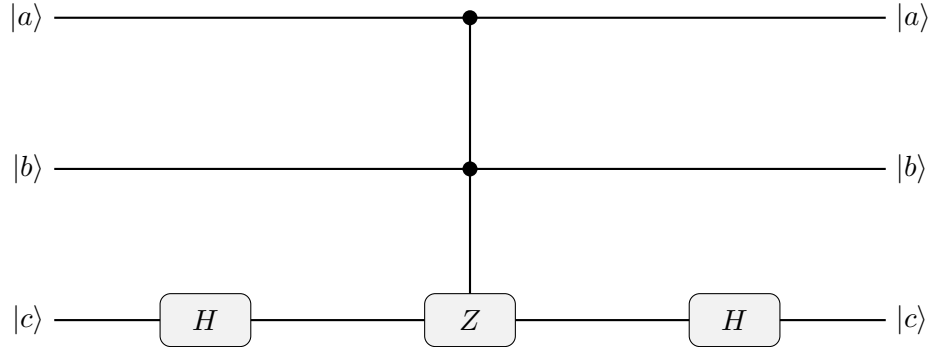
we obtain

$$F_{\text{loc}} U F_{\text{loc}}^{-1} = \text{CCZ}.$$

Thus

$$\text{CCZ}|a, b, c\rangle = (-1)^{abc}|a, b, c\rangle.$$

8.1 Circuit for the Fourier Representation of Toffoli



9 Fourier Tower Representation of the Fredkin Gate

The Fredkin gate is the controlled-SWAP gate:

$$U = \text{CSWAP}.$$

It is defined by

$$\text{CSWAP}|a, b, c\rangle = |a, b, c\rangle \quad \text{if } a = 0,$$

and

$$\text{CSWAP}|1, b, c\rangle = |1, c, b\rangle.$$

Equivalently,

$$\text{CSWAP} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \text{SWAP}.$$

The target symmetry group is

$$S_2 \cong \mathbb{Z}_2.$$

The two-qubit target space decomposes as

$$\mathcal{H}_{\text{target}} = \mathcal{H}_{\text{sym}} \oplus \mathcal{H}_{\text{antisym}},$$

where

$$\mathcal{H}_{\text{sym}} = \text{span} \left\{ |00\rangle, |11\rangle, \frac{|01\rangle + |10\rangle}{\sqrt{2}} \right\},$$

and

$$\mathcal{H}_{\text{antisym}} = \text{span} \left\{ \frac{|01\rangle - |10\rangle}{\sqrt{2}} \right\}.$$

The SWAP operator acts as $+1$ on the symmetric sector and -1 on the antisymmetric sector. Therefore the Fourier representation of Fredkin is

$$\tilde{U} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \text{diag}(1, 1, 1, -1).$$

9.1 Circuit for Fredkin

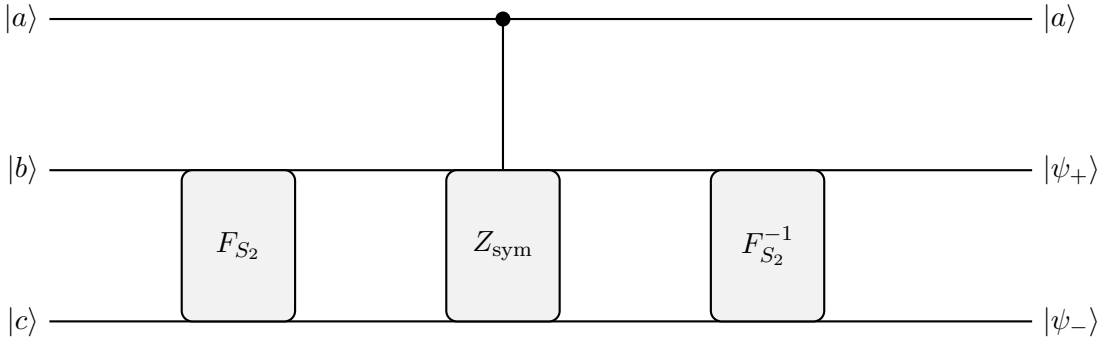


Figure 2: Corrected Fourier Tower representation of the Fredkin gate. The gates F_{S_2} , Z_{sym} , and $F_{S_2}^{-1}$ act on the two target wires together. The controlled phase acts on the symmetry-sector register, not outside the circuit wires.

This example illustrates

permutation symmetry $\longrightarrow$ phase separation of symmetry sectors.

The quantum information viewpoint follows the standard framework of [5, 8].

10 Conclusion

The examples above show concretely how the Fourier Tower works.

For S_3 , nonabelianity appears as a controlled permutation of Fourier modes.

For S_4 , the tower has several coupled layers:

$$V_4, \quad \mathbb{Z}_3, \quad \mathbb{Z}_2.$$

For D_4 , all quotient layers are binary, so the local Fourier transform is a product of Hadamard gates.

For Q_8 and H_p , nonabelianity appears through cocycle-induced phase twisting.

For Toffoli, Fourier conjugation turns a controlled translation into a controlled phase.

For Fredkin, Fourier conjugation resolves permutation symmetry into symmetric and antisymmetric sectors.

Together, these examples illustrate the principle

nonabelianity = controlled interaction between abelian Fourier layers.

A Physical Origin of the Heisenberg Cocycle

The connection with Weyl operators and quantum mechanics is classical [4, 7].

A.1 The Continuous Heisenberg Relation

In ordinary quantum mechanics, the position and momentum operators

$$\hat{x} \quad \text{and} \quad \hat{p}$$

satisfy

$$[\hat{x}, \hat{p}] = i\hbar I.$$

Equivalently,

$$\hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar I.$$

Thus position and momentum do not commute. The failure of commutativity is measured by a central scalar term.

This is the continuous origin of the Heisenberg group.

A.2 Weyl Operators

Instead of working directly with $\hat{x}$ and $\hat{p}$, one introduces the Weyl operators

$$U(a) = e^{ia\hat{x}}, \quad V(b) = e^{ib\hat{p}}.$$

They satisfy

$$U(a)V(b) = e^{iab\hbar}V(b)U(a).$$

The factor

$$e^{iab\hbar}$$

is a phase cocycle.

This is the physical analogue of the finite cocycle

$$\omega((a, b), (a', b')) = ab'$$

appearing in the finite Heisenberg group.

Thus the finite Heisenberg Fourier Tower may be viewed as a discrete quantum version of the passage

noncommuting translations $\longrightarrow$ central phase correction.

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